A Tiered Peer-Critique Architecture for AI Answers

Every major lab's models sit on a panel and argue it out. A hub that never writes asks the questions.

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Nothing picks a model.	**Nothing assigns the work.**	**The hub never writes.**

Summary

No single AI model is best at everything, and none is reliable at knowing its own weak spots. Convening a panel is now a shipped feature. This proposal is about what happens after the panel is convened: the answer has to earn its release.

Every request runs through the current publicly available models from every major lab, always the latest live version of each, sorted into class tiers. The models break the request down, bid on its parts, challenge each other's drafts, and revise until they agree. A dedicated hub that never writes an answer questions the result until nothing is left open, releases it with a note on where the panel split, and keeps a running catalogue of who was challenged, who was corrected, and who was later proven wrong, so the same class of mistake is caught sooner next time.

Where an answer can be checked, this should beat the best single model by a measurable margin. Where it can't, the design says so rather than pretending otherwise. A compute-matched pilot that could show either result is specified at the end.

What this proposal adds

Debate between models, councils of models, and judges of models all exist; the prior-work section says where. These six elements are what the design contributes, each named against its nearest neighbor.

1. **A hub that never writes**

   The questioner is separated from the authors. It has no answer of its own to defend, picks no winner, and produces no synthesis; it asks until nothing is open, within a budget, and enforces checks it does not need to understand.

   Nearest: the judge in debate [1, 3, 4] and the questioning judge in consultancy [2, 4], both of which decide.

2. **Bidding instead of routing**

   No component picks a model. Models bid for parts of the request as equals and hand them off mid-stream; critique, not a router, sorts good bids from bad.

   Nearest: learned routers [16], confidence-aware routing [21], and the contract net [17], all of which award the work.

3. **Seat-anonymous, evidence-required critique with a designated dissenter**

   Drafts carry a seat number, not a lab name; every challenge must point at evidence; one seat per round is assigned to attack the consensus.

   Nearest: personas and roles in multi-agent debate [8], and the findings on authority bias [13], debate hacking [10], and persuasion [20] that motivate the rule.

4 **Release conditions instead of a round count**

Nothing ships until the panel agrees, the hub has no open questions, and every verifiable check passes. Deadlocks escalate to the next tier or ship with the dispute flagged to the user.

Nearest: fixed rounds [7] and fixed layers [9], which stop on a count.

5 **A persistent, version-stamped failure catalogue**

The hub keeps a record across users of who was challenged, corrected, and later shown wrong, stamped with the model version, and draws its patterned questions from it.

Nearest: static evaluation harnesses, and the per-user memory in council products, which remember the user rather than the errors.

6 **Cross-lab tiers with weighted agreement**

Tiers are matched by model class across labs. Agreement between a lab's light and frontier models is discounted because the light one is usually distilled from the other, and the hub must match the top tier in play.

Nearest: mixture-of-agents layers [9], and the capability-gap results for weak judges [4, 5].

The panel

Membership. The major labs' hosted models by default, always the latest live version of each. Open-weight models (Llama, Mistral, DeepSeek, Qwen and the like) are available as an add-on.

Tiers. Models are grouped by class, and each tier pulls the equivalent model from every lab. The tier a user pays for decides which classes work the request. Every tier runs the same process; cheaper tiers simply run it with cheaper models.



At the god tier, the frontier models choose the final answer by looking at where the tiers agree. That agreement is weighted rather than counted one model, one vote: a lab's light model is usually distilled from its frontier model, so the two tend to be wrong in the same places, and agreement between them adds little. The lighter tiers earn their seat a different way, as cheap coverage. They catch omissions, offer alternative readings, and give a fast first pass. When a light model disagrees with a frontier model, it is more often the light model that is wrong, and the split is still recorded.

In a perfect world every request would run at the god tier. The tiers exist because that isn't affordable for the everyday user today.

The hub

> A dedicated model that never sits on the panel, so it has no answer of its own to defend. It does not choose a model, edit the answer, or define it. **It asks.**

What it asks

What is the source. Why did seat B disagree. What happens if X. It keeps asking until nothing is left open, within a question budget, so interrogation neither loops forever nor turns into a rubber stamp.

What it must be

At least the class of the highest tier in play. A questioner weaker than the panel it examines gets talked past by confident wrong answers, which is what the weak-judge experiments found once the capability gap widens [4, 5]. Two things narrow that gap: the hub can enforce checks it doesn't need to understand (run the code, fetch the source, recompute the figure), the way a clerk enforces procedure, and the catalogue hands it patterned questions drawn from past failures.

Where it comes from

Every lab has a model on the panel, so the hub is either supplied by a rotating lab or trained for the role. Either way, it keeps the catalogue.

How a request moves

The same six steps run in every tier.

1. **Decomposition by the panel**
   The panel proposes a breakdown of the request into components and critiques it the same way it critiques answers. At the god tier the frontier models propose the breakdown and the lighter tiers work from it, so the request isn't debated twice. The hub does not decompose.

2. **Bidding, not assignment**
   Each model puts up which components it is prepared to answer. Bids are equal: no central picker, no ranking by reputation. Models can hand components to each other mid-stream, and one model may claim the whole request.

3. **Mandatory critique**
   Every draft, whole or partial, is challenged by the other models before it goes anywhere. Drafts are attributed by seat, not by name, so a challenge targets the content rather than the label on it, and every challenge has to point at evidence. Critique is what separates good bids from bad ones.

4. **Iterative molding**
   The answer is revised across rounds, by different models, until the panel agrees it is ready to submit. Each tier has a round budget.

5. **Interrogation**
   The hub questions the submitted answer until it has nothing left to ask.

6. **Release**
   The user receives the answer, a note on where the panel split, and the full transcript on demand.

Release conditions

All three are required before an answer leaves the system.

- ✓ **The panel agrees**: within the tier on single-tier plans, across tiers at the god tier.
- ✓ **The hub has no open questions left.**
- ✓ **Every verifiable check passes**: code runs, sources are found, numbers recompute.

> **Deadlock.** If the panel can't agree, or a hub question can't be answered because no source exists, within the round or time limit, lower tiers may escalate to the next tier up where the plan allows. At the god tier there is nothing above, so the best draft ships with the dispute flagged to the user.

Guarding against shared blind spots

Because the models trained on much the same data, unanimous agreement can be unanimous error [11, 12].

Version 1

Cross-tier agreement plus the verifiable checks.

The fuller version

A designated dissenter each round, whose only job is to attack the consensus, and a rule that no factual claim passes without a source or a check attached.

The catalogue

The hub records who bid on what, who was challenged on what, where the panel split, what was corrected, and what was later found wrong. It uses that record to spot the same class of inaccuracy sooner and to sharpen its own questions.

Two layers

A global record of patterns across all users, and a private per-user layer.

Ground truth

The catalogue logs an error, rather than a disagreement, only once something confirms who was right: user feedback and corrections, automated checks, and outcomes that surface later (a prediction, a figure, or a diagnosis confirmed down the line). All three feed the record.

Version stamps

Because the panel always runs the latest live version of each model, every entry is stamped with the model version it describes. A pattern that belonged to last quarter's model is evidence about this quarter's, not a verdict.

Retention

Transcripts on demand plus a per-user layer means storing a great deal. The retention policy is a day-one decision, not a later one.

Where the gain lives

Where an answer can be checked (math, code, lookups, data work), the panel should beat the best single model by a measurable margin, and the verifiable-checks rule is what makes the gain stick. On open-ended judgment, writing, and taste, the evidence points the other way: consensus flattens answers, and models tend to cave under critique even when they were right. There, the panel's honest output is the split note, not a better paragraph.

The cost is real. A panel of N models over R rounds spends tens of times the tokens of a single answer and takes minutes rather than seconds. That is acceptable at the god tier and needs a strict round budget at the light tier.

One benchmark rule applies before anything is priced: each tier is measured against a single frontier-class model, not against a single model

of its own class. A panel of light-class models will not generally beat one frontier model, and a tier that can't should be priced and described accordingly.

Prior work

This design sits inside two lines of research and borrows from a third. Naming them is meant to show where the proposal starts, not to claim any of it.

Debate as oversight

Irving, Christiano and Amodei proposed in 2018 that two AIs argue before a judge, on the reasoning that a lie is harder to defend than to refute [1]. Experiments with human judges followed [2], then with model judges: Khan and colleagues found that more persuasive debaters make judges more accurate [3], and Kenton and colleagues put weak model judges over strong debaters across extractive QA, math and code, finding that debate beats a single consultant, that its edge over the judge answering alone is mixed once there is no information asymmetry, and that stronger debaters help only modestly [4]. Later work has modeled how oversight scales with the capability gap [5] and studied collaborative rather than adversarial resolution of disagreements [6]. The hub here is a descendant of the judge with one role removed: it never decides for a side and never writes an answer.

Multi-agent debate for accuracy

Du and colleagues showed in 2023 that several instances of a model answering, reading each other, and revising over fixed rounds improves math and factual accuracy [7]. Variants added personas and assigned roles [8], and mixture-of-agents arranged models from different labs in layers with an aggregator writing the final answer [9]. Recent work goes after the failure modes: debate hacking, where debaters mislead the judge with overconfident claims and end up below single-agent baselines, answered with a non-zero-sum protocol [10]; homogeneous reasoning paths and deadlock [11]; the roles of confidence and diversity [12]; and authority bias in multi-agent evaluation [13]. The panel here is the collaborative, cross-lab branch of this line, with the aggregator removed.

Judges, routers and aggregators

Using a model as a judge is standard practice and has known biases, including a preference for a model's own outputs [14, 15]. Routers such as RouteLLM learn to send each request to one model [16], and commercial gateways do the same in production. Bidding for work goes back to the contract net of 1980 [17]. This design uses none of them to pick a model: bidding replaces routing, critique replaces the aggregator, and a questioner replaces the judge.

What the field has found wrong with all of this

- A single model with a strong prompt matches the best discussion methods on a wide range of reasoning tasks [18]; debate strategies do not reliably beat self-consistency and ensembling once cost is counted [19]; and under equal thinking-token budgets, single agents have been reported to beat multi-agent systems on multi-hop reasoning [22]. This is why the pilot below is compute-matched: the panel is compared against one frontier model given the same token budget, not against one pass.
- Judges get talked past [4, 10], defer to authority [13], and are vulnerable to persuasive but wrong reasoning [20, 23]. This is why critique is seat-anonymous and evidence-required, why a dissenter is designated, and why the hub enforces checks it does not need to understand.
- Models trained on the same data share blind spots, so agreement is weak evidence [11, 12]. This is why agreement is weighted across labs and tiers rather than counted, and why nothing ships without a verifiable check where one exists.

Three recent results point the same way as the design. The one lever that made debate competitive in Smit and colleagues' benchmark was tuning how readily agents agree [19], which is the quantity the release gate and the designated dissenter control directly. A 2026 study of debate for data cleaning reports that self-verification with identical tools failed, and that a separate critic grounded in code execution with evidence-gated generation was the first debate configuration to significantly beat a single agent on a generative task [24]: a separate questioner, mechanical checks, and an evidence rule are three of this design's six elements. And heterogeneous panels, models from different labs rather than copies of one, have been reported to show honest gains and resilience under adversarial peers [25], which is the case for filling every seat from a different lab.

What's different

Hub that never writes

Nearest: Judge in debate [1, 3, 4]; questioning judge in consultancy [2, 4]

Here: No verdict between sides and no authored answer. Release is by exhaustion of questions within a budget, plus checks the hub enforces without needing to understand them.

Bidding, not assignment

Nearest: Learned routers [16]; confidence-aware routing [21]; contract net [17]

Here: No picker and no award. Models bid for components as equals and hand them off mid-stream; critique sorts good bids from bad.

Seat-anonymous, evidence-required critique, with a dissenter

Nearest: Roles in multi-agent debate [8]; authority bias [13]; debate hacking [10]

Here: Anonymity removes the label, the evidence rule removes bare assertion, and the dissenter makes disagreement structural rather than hoped for.

Release conditions

Nearest: Fixed rounds [7]; fixed layers [9]

Here: Three conjunctive gates (agreement, no open questions, checks pass), deadlock escalation across tiers, and disputes surfaced to the user rather than smoothed over.

Failure catalogue

Nearest: Static evaluation harnesses; per-user memory in council products

Here: Memory of who was wrong and on what, shared across users, stamped with the model version, and fed back into the hub's questions.

Cross-lab tiers, weighted agreement

Nearest: Mixture-of-agents layers [9]; weak-judge results [4, 5]

Here: Tiers matched by class across labs; agreement within a lab's model family discounted; the hub required to match the top tier in play.

What this proposal does not claim

- That debate, councils, or model judges are new. They are not.
- That the panel beats a single model on open-ended work. The evidence says it flattens such answers, and the design's output there is the split note.
- That shared blind spots are solved. They are discounted and checked, not removed.
- That this is cheaper. It is slower and costs more, and the pilot counts that against it.

Products and systems today

The pieces exist separately, and since early 2026 the panel itself is a shipped product. No known system combines tiers, pickerless bidding, a hub that only questions, and a cross-user error catalogue.

Multi-agent debate (research since 2023)

Several model instances answer, read each other's answers, and revise over a fixed number of rounds, with measurable gains on math and factual questions. Falls short: usually one model talking to copies of itself, no tiers, no neutral hub, no memory, and it stops on a round count rather than on agreement.

LLM councils (open source, November 2025)

Several models answer through one API broker, anonymously rank each other's answers, and a chairman model writes the final response.

Falls short: the chairman authors, which this design forbids; one round; nothing is remembered.

Council products (2026)

Perplexity's Model Council (February 2026, opened to Pro subscribers in July) sends one prompt to between two and eight models and has a chair synthesize them, flagging where they converge or clash. Council AI runs twenty-plus models from nine labs to a moderator's verdict and keeps a per-user memory; Roundtable runs three models and a moderator. Falls short: in every case a synthesizer authors the answer; there is no critique-and-revise loop that ends on agreement, no release gate, and the only memory is of the user, not of who was wrong.

Mixture-of-agents

Layered aggregation, where each layer's models read the previous layer's outputs and an aggregator produces the answer. Falls short: the layers loosely resemble tiers, but aggregators author, there is no critique dialogue, and there is no memory.

Contract-net bidding (1980)

The ancestor of step 2: tasks are announced, agents bid. Falls short: a manager awards the task. Here nothing awards, and critique does the sorting.

Routers and orchestrators

Pick one model, or assign work from the top. The opposite of this design, on purpose.

The catalogue has no close analog. The nearest things are evaluation harnesses (static test sets) and per-app lessons-learned memories; neither feeds a live critique loop across users.

Known risks

- **Incentives and terms of service.** A catalogue that scores each lab's models against the others is a public scorecard, and some labs restrict using their outputs to evaluate or improve competitors. Expect friction, and expect the choice of who supplies the hub to matter.
- **The question budget is a tuning problem.** Too small and the hub is a rubber stamp; too large and the god tier never ships.
- **Deference.** Even with seats instead of names, models can often recognize each other's style. Anonymized critique reduces caving; it doesn't eliminate it. Kraidia and colleagues study persuasion-driven adversarial influence in multi-agent debate [20] and are the reference for this risk; the designated dissenter and the verifiable-checks rule are the answer to it.
- **Moving targets.** Always-latest versions mean behavior shifts without notice. The version stamps in the catalogue are the defense, and they have to be checked, not assumed.

Pilot design

Everything above rests on two claims that can be tested cheaply: that the panel beats the best single model where answers can be checked, and that the hub catches what the panel misses. The pilot is built to be argued with, so the metrics are fixed before the first run.

Tasks

64 freshly written items across three checkable types: 24 arithmetic and word problems with a numeric answer, 20 data-table questions with a computed answer, and 20 short coding tasks with hidden unit tests. The answers are produced by the same code that writes the prompts, so the key is right by construction, and no public benchmark item is used, so contamination is not in play. Factual lookups that must cite a resolvable source are deferred until web-enabled seats are in place.

Conditions

(a) The best single frontier model, one pass. (b) The same model given the panel's token budget: k samples with a majority vote (or a self-review pick for code), then a self-critique pass, where k is set per task type from the tokens the panel actually spent. (c) The panel with the hub, at the mid tier, running the prototype's six steps prompt for prompt. (d) The panel with the hub removed, to isolate what the hub adds.

Metrics

Accuracy per condition with confidence intervals; tokens and wall-clock per item; the hub's catch rate (answers the panel agreed on, the hub

sent back, and that were in fact wrong) and its false-alarm rate; and the deadlock rate.

What would falsify the design

If (c) does not beat (b) on the checkable tasks, the panel is not earning its cost. If (c) does not beat (d), the hub is decoration. Either result is reported as found.

Status

The harness, the task set, and the report generator are in the repository that accompanies this draft, tested end to end against a mock backend. Results against live models will be added in the next version, whichever way they come out.

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Entries without an author list or identifier are cited by title and year; the working draft will fill them in before any formal submission.

Working draft v1.1, September 2026. Drafted with Claude (Anthropic). Text licensed CC BY 4.0; the accompanying prototype is MIT licensed.

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